

Ethics Scenario Assignment Part 2

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Part A:

I'd start by making sure Neville knows he's not in trouble and that his work is valued: "Neville, your accuracy on the QC line is excellent. We just need to figure out the sampling pace, since about every third sample is being missed, and I want to make sure the process works for you."

I wouldn't ask about disabilities—that's private—but I'd focus on what's making the task difficult: Is it belt speed, the motion of picking beans, or the station setup? My experience managing teams taught me that people usually know what the barrier is if you ask in a supportive way.

1. Adjust the Sampling Process: If the belt is too fast, we could slow it slightly during Neville's shifts or widen the sampling interval. As long as QC standards are met, this keeps Neville in his role and accuracy high.
2. Modify the Workstation: Adding a mechanical diverter or chute could separate every 30th bean automatically, so Neville inspects it without grabbing from the moving belt. This not only helps Neville but could benefit anyone at that station. Framing it as a process improvement avoids stigma and helps with buy-in.

Part B: Reflection

Filling out the UW DO-IT checklist for our lab space made me realize how many "invisible" barriers exist. As a senior student, I've seen how easy it is to focus only on chemistry or math and completely ignore whether a person can reach the equipment or navigate the aisles safely.

I feel fairly comfortable thinking about these impacts. Being a bit older than my peers, I'm already aware that everyone's physical and mental processing speeds vary, and I believe advocating for a more accessible workplace just makes for a better engineering team overall. I've worked with classmates who had invisible disabilities, and I know how much it means to have someone on the team who considers those needs.

The exercise prompt - Yes, this was useful. In engineering, we are taught to optimize systems. If a system (like a lab or a factory) doesn't work for 27% of the population, then it isn't optimized. This exercise really shifted my mindset from "fixing the person" to "fixing the design." I started looking around campus with new eyes after this—seeing curb cuts, automatic doors, and even font choices as part of design optimization.

Workplace assessment utility:

I think this would be very useful in a professional setting. It's much cheaper to catch an accessibility issue during the design phase of a factory like Bertie Botts' than it is to retroactively fix it after someone gets hurt or a line fails. I've seen companies scramble to add last-minute fixes, and it's never as smooth as doing it right from the start.

Relating the checklist to the scenario: the checklist likely mentions "reach ranges" or "manual dexterity requirements" for equipment. This relates directly to Neville's struggle to pick beans off the belt—the "interface" between the human and the machine is where the failure is happening. I always think about how even small design tweaks can prevent big problems.

Missing assessments: I think the checklist could include more about "cognitive load" or how instructions are delivered. Most people don't recognize the toll of age on mental speed and the ability to process new information. We know the science of it, but the application to school and work has lagged way behind.